

### **Subject Description Form**

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| <b>Subject Code</b>                                   | BSE5411                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Subject Title</b>                                  | Building Carbon Footprint Assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Level</b>                                          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"><li>1. To gain an understanding of climate change and national and international gas emissions targets and reporting requirements.</li><li>2. To develop skills in carrying out carbon impact assessment for buildings.</li><li>3. To gain an understanding of how to achieve a reduction target in greenhouse gas emissions for building construction, operation and maintenance.</li><li>4. To formulate low-carbon strategies for building designs.</li><li>5. To acquire basic skills on conducting carbon reporting assessments for companies and organizations based on international and local carbon reporting guidelines.</li></ol>                                                                                                                                                                                                                                               |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"><li>a. understand the impacts of climate change, and the international and local mitigation strategies and policies;</li><li>b. understand the impacts of buildings on carbon emissions;</li><li>c. appreciate the concept of embodied carbon and the carbon footprint of materials and products (using both a process based end to end lifecycle approach and input-output techniques);</li><li>d. estimate the embodied energy and carbon footprints of the building material use, and energy use due to building operation;</li><li>e. identify and prioritize appropriate cost effective design and operating strategies to reduce greenhouse gas emissions in a variety of scenarios; and</li><li>f. apply appropriate quantification methodologies for greenhouse gas emissions mitigation and reporting.</li></ol> |

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| <b>Subject Synopsis/<br/>Indicative<br/>Syllabus</b> | <p><b>Climate Change and Potential International and Local Challenges:</b> climate change; international collaborative efforts, Kyoto Protocol, impact of buildings.</p> <p><b>Major Carbon Reduction Policies:</b> carbon tax, emission trading, energy saving labels etc.</p> <p><b>Greenhouse Gas:</b> types and sources, emissions and removals from building activities, accounting and reporting for buildings.</p> <p><b>Low Carbon Building Design and Building Carbon Footprint Assessment:</b> embodied energy and carbon emissions for material use and operations in buildings, carbon-cost curves, achieving low carbon design through the application of both conventional and state-of-the-art mitigation strategies with the lifecycle cost considerations.</p> <p><b>Building Carbon Audit and Reporting:</b> purposes and functions, overseas and local developments, regulatory control or voluntary participation, carbon accounting protocols (e.g. ISO14064), product carbon accounting standards (e.g. PAS2060:2010).</p> <p><b>Quantification methodologies:</b> physical and operational boundaries of buildings, scopes of carbon emissions and removals from buildings, data requirements and analyses.</p> |
| <b>Teaching/Learning<br/>Methodology</b>             | <ol style="list-style-type: none"> <li>Lectures will be delivered to introduce the concept relating to climate change, carbon impact assessment, auditing etc.</li> <li>Workshops will be organized to enable the students to apply what they learn in lectures.</li> <li>Students will be required to present their solutions and findings obtained from workshop to their peers and lecturers in workshops.</li> <li>Final examination will be held to examine the learning outcomes of students.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

